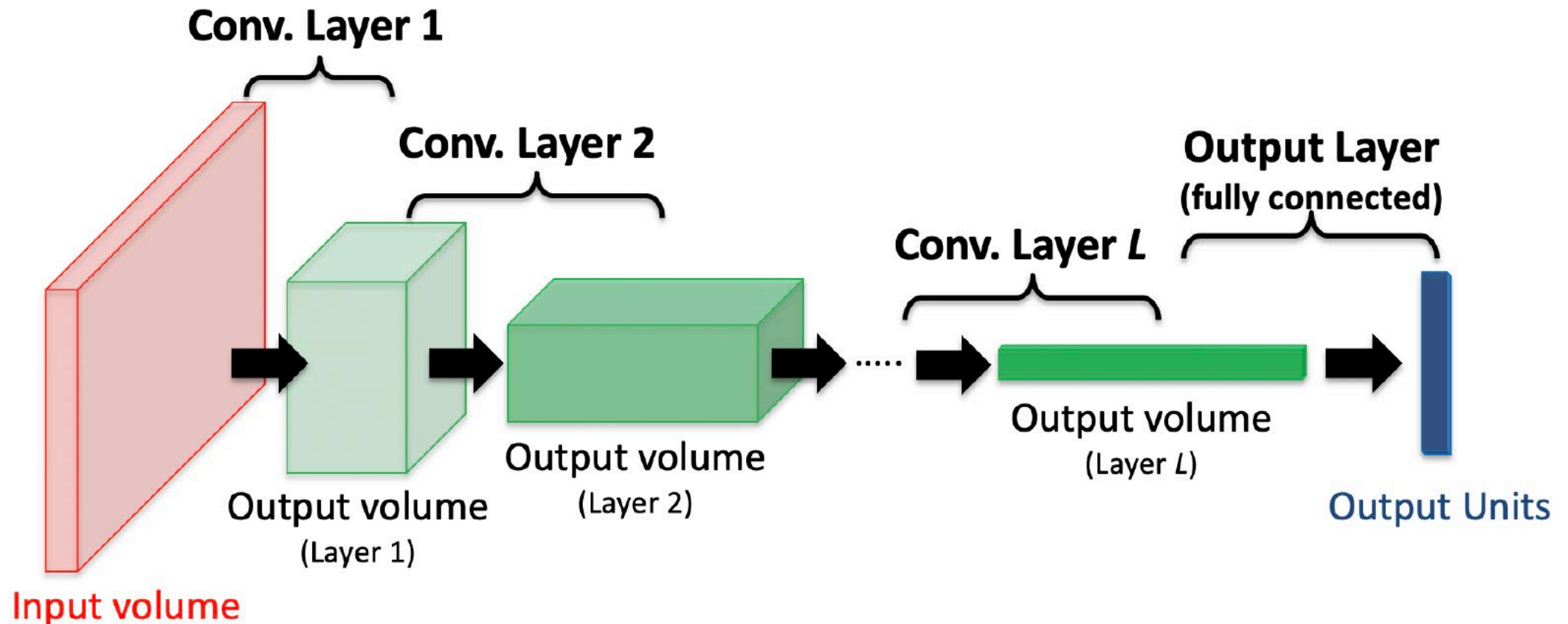


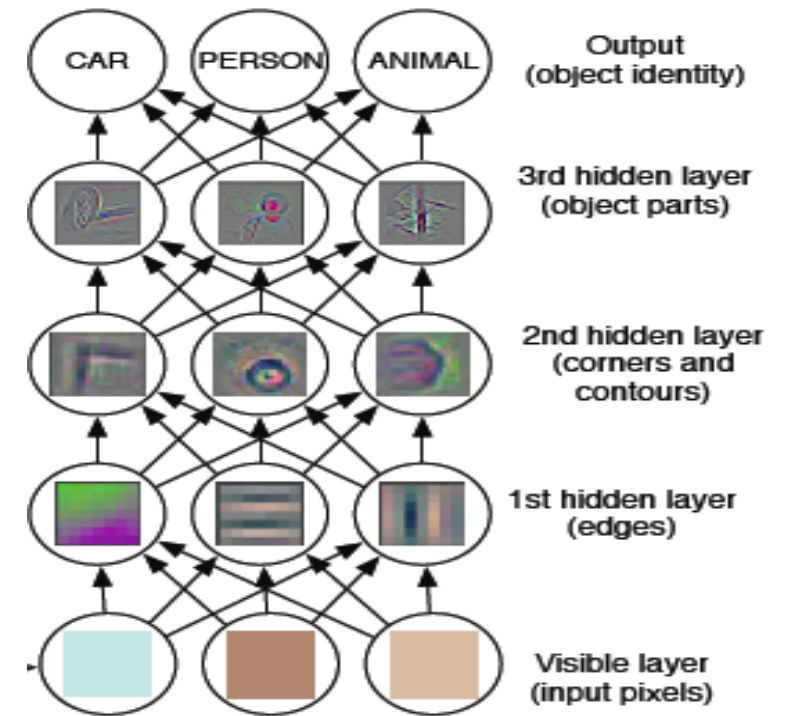
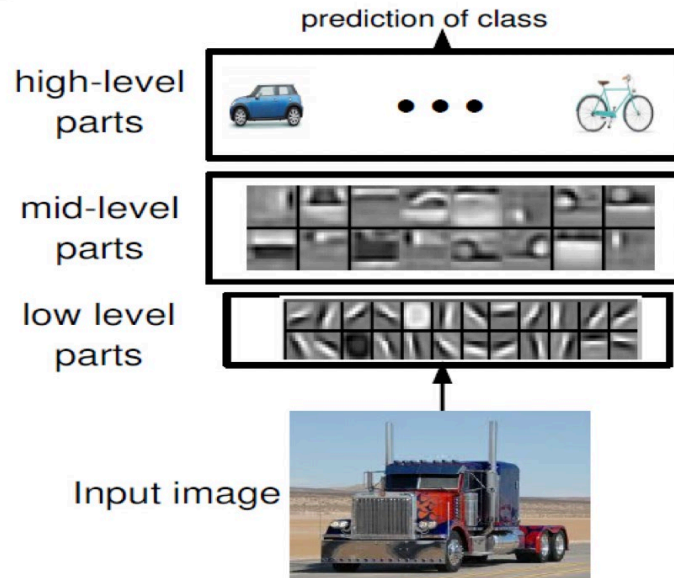
Convolutional Neural Networks

- CNN = NN has at least one convolutional layer

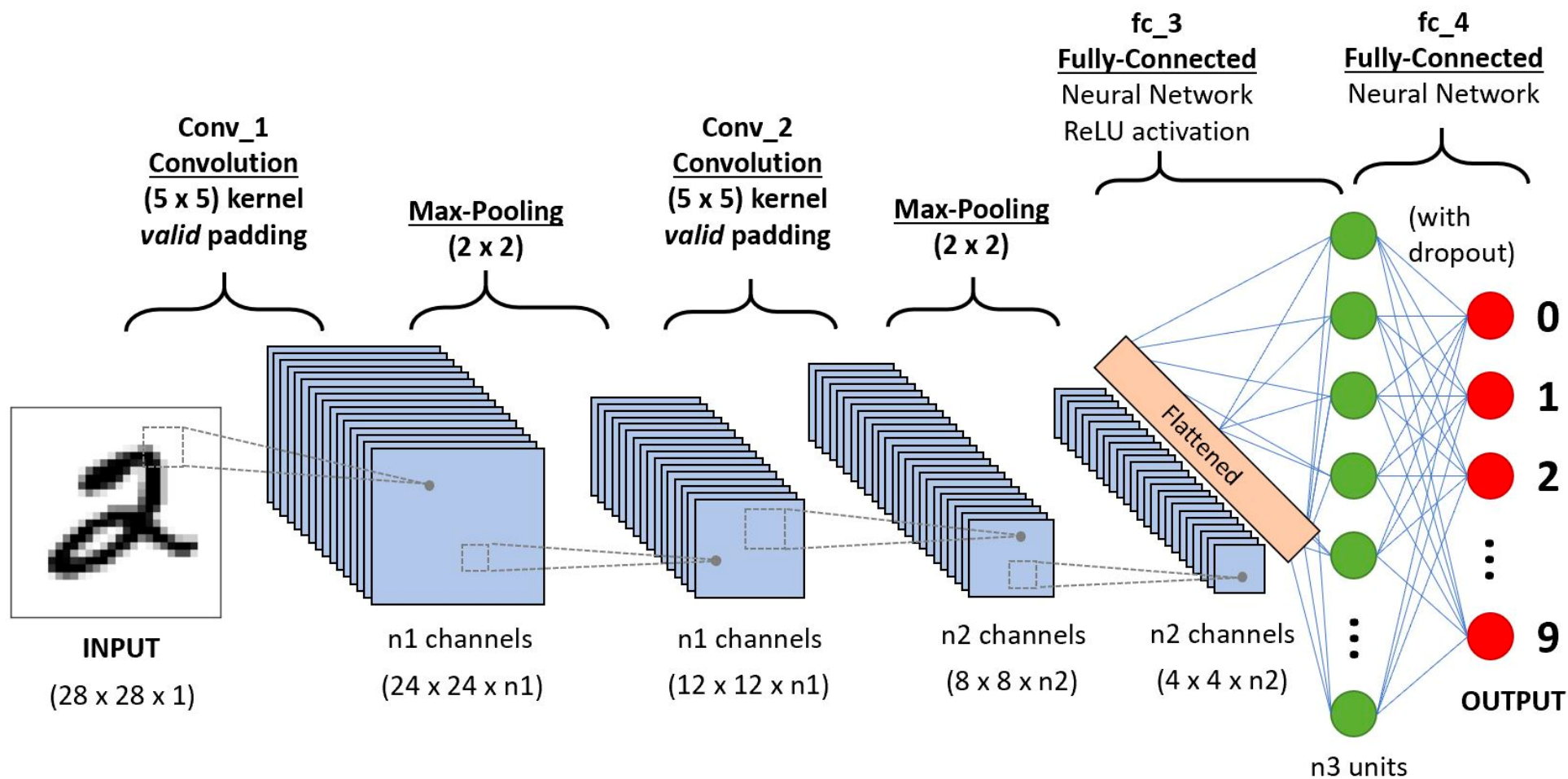


Importance of # of layers

- ❑ Models goes deeper (# of layers increases), CNNs tend to learn “better” representation
- ❑ Why? The higher layer is able to learn from a larger receptive field.
- ❑ Examples
 - ❑ In 1990s: LeNet has 8 layers
 - ❑ Now: it is common to have 100+ layers (DenseNet, resNet)



Convolutional NN for Image Recognition: Example



An example CNN for hand-written digit recognition